

A multi-architecture study of specificity refinement and false-positive mechanism analysis in prostate MRI

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Abstract

Objectives. To characterize residual false positives in prostate MRI detection, and to evaluate a lightweight post-hoc refinement head for case-level specificity.

Materials and Methods. This retrospective study used PI-CAI (5-fold cross-validation) and Prostate158 ($n = 158$; external). A context-aware evidence head and an 89,216-parameter refinement head were trained on a frozen detection backbone; the evidence head was also trained on four further backbones (bare nnU-Net, bare U-Net, bare Mamba, MIGF-Mamba). For each false-positive region, T2-weighted, apparent-diffusion-coefficient, and high-b-value contrast ratios versus peri-lesional rings were compared against ground-truth lesions and contralateral benign regions.

Results. False positives were closer to true cancers than to benign tissue in evidence and raw T2-weighted and apparent-diffusion-coefficient contrast, reproducing 35/35 across five architectures (Cohen’s d 1.10; FP/benign evidence ratio 2.38x) and 105/105 across modality-perturbation scenarios. On PI-CAI fold-0, refinement raised case-level specificity from 0.469 ± 0.181 to 0.549 ± 0.132 (+17.2%) at preserved sensitivity (0.943 ± 0.021); 5-fold cross-validation showed fold-conditional behavior (9/15 observations positive; range -22% to +28%). On Prostate158, both models saturated (McNemar pooled $p = 0.69$), while the false-positive contrast-matching finding replicated.

Conclusion. Residual false positives are contrast-matched to cancer (sharing raw imaging features rather than histologically confirmed mimicry), reproducing across five architectures – a data-level imaging property, not model-specific artifacts; post-hoc refinement adds practical specificity in-domain but is fold-conditional.

Keywords: Prostate cancer; Magnetic resonance imaging; Deep learning; False positive reactions; Image interpretation, computer-assisted

Key Points

- **Question.** Why do false positives persist in prostate MRI AI, and can a lightweight post-hoc head improve case-level specificity without sensitivity loss?
- **Findings.** False positives resembled true cancers more than benign tissue (35/35 across five architectures); a post-hoc head improved fold-0 specificity by 17.2%.

Clinical Relevance Statement

False positives from prostate MRI AI detectors are radiologically interpretable as cancer-resembling regions, not arbitrary model artifacts; this finding reproduces across five distinct architectures and informs how AI-flagged regions should be adjudicated.

Introduction

Clinically significant prostate cancer (csPCa) is diagnosed from multiparametric magnetic resonance imaging (MRI) followed by targeted biopsy of suspicious lesions. Prostate MRI-based artificial intelligence (AI) systems have advanced substantially on benchmark datasets such as PI-CAI [1], yet clinical translation is held back by high false-positive rates: regions flagged by AI often do not correspond to biopsy-confirmed cancer, triggering unnecessary follow-up imaging and biopsies [2, 3]. Existing AI pipelines are overwhelmingly tuned to maximize sensitivity, while specificity receives comparatively little attention and the nature of residual false positives is rarely characterized in a way that informs clinical action on AI output.

Two gaps motivate this work. First, improving case-level specificity of a deployed AI detection model typically requires full retraining; simple post-hoc recalibration (temperature or Platt scaling) cannot alter case-level specificity beyond what threshold tuning achieves, since these methods preserve ranking. Lightweight post-hoc specificity refinement on a frozen backbone that preserves sensitivity is therefore under-explored in prostate MRI. Second, false positives are usually summarized as a single error rate; their evidence structure — whether they resemble true cancers in raw imaging contrast or are arbitrary model artifacts — is rarely disentangled. Without that distinction, recommendations for radiologist adjudication and triage remain speculative.

Here we first quantify the evidence structure of residual false positives. Using a context-aware evidence head and direct raw-imaging contrast on T2-weighted and apparent-diffusion-coefficient channels, false positives from a frozen csPCa detection backbone [4] are consistently closer to true cancers than to benign prostatic tissue — a direction reproducing across PI-CAI 5-fold cross-validation, seven modality-perturbation scenarios, and five distinct backbone architectures. As an implementation this mechanism analysis explains, we introduce a lightweight (89 K-parameter) post-hoc specificity refinement head; on PI-CAI fold-0 it increases case-level specificity by 17.2% at preserved sensitivity, with 5-fold cross-validation revealing fold-conditional magnitude. We replicate both analyses on Prostate158 [5] and report what works, where center-specific effects limit the claim, and what this implies for AI-assisted prostate MRI.

Materials and Methods

This study analyzed de-identified multiparametric prostate MRI from two publicly available, institutionally approved cohorts (PI-CAI and Prostate158) under their respective data-use licenses;

no additional institutional review board approval was required and the requirement for informed consent was waived.

3.1 Datasets

This retrospective study used two public prostate multiparametric MRI datasets. The PI-CAI Public Training and Development Dataset [1] (1500 multi-vendor studies from 1476 patients across three Dutch centres; 425 biopsy-confirmed csPCa, 1075 pathology-negative) served as the primary cohort; we used the official 5-fold split, restricting method development and primary evaluation to fold-0 (1200 training / 300 validation). Prostate158 [5] (158 studies with biopsy-confirmed labels) served as the external cohort. Both datasets provide axial T2-weighted, high-b-value diffusion-weighted, and apparent-diffusion-coefficient sequences. Preprocessing — axial resampling to $0.5 \times 0.5 \times 3.0$ mm, gland-centered cropping to $128 \times 128 \times 32$ voxels, and per-channel normalization — followed our prior work [4] and was identical across datasets.

3.2 Frozen detection backbone

All experiments used a frozen multi-modal csPCa detection backbone from our earlier work [4]. The backbone accepts three co-registered channels (T2-weighted, high-b-value DWI, ADC) at $128 \times 128 \times 32$ voxels; each modality has its own four-level encoder ($3 \times 3 \times 3$ convolution, downsampling 2 per level). Adaptive gating fuses encoders at the lowest-resolution stage and feeds a shared three-stage decoder with dec1, dec2, dec3 outputs at 32/64/128 channels and full/half/quarter resolution; a $1 \times 1 \times 1$ convolution returns per-voxel logits. Each input channel was zeroed with probability 0.3 per sample during backbone training; dropout was disabled at inference. We reused five-seed checkpoints (42, 123, 456, 789, 1024) for the A2 variant (gating + modality dropout, without deep supervision) unchanged.

We selected this backbone for three reasons: (i) **modality isolation** via per-modality encoders enables deterministic per-channel attribution required by the evidence-grounded analyses in §3.4; (ii) **clinical-grade efficiency** with 9.45 M parameters and 8.5 ms single-volume inference latency leaves compute headroom for downstream uncertainty and risk-stratification pipelines; (iii) **measurable-gain headroom** since the A2 variant’s 5-fold mean case-level specificity (0.46) avoids ceiling effects that would compress observable post-hoc refinement gains.

3.3 Post-hoc specificity refinement head (P2a)

The refinement head takes the backbone’s decoder features at three resolutions (dec1/dec2/dec3; 32/64/128 channels). The dec2 and dec3 streams are projected to 32 channels and trilinearly upsampled to dec1; the three streams are concatenated (96 channels), passed through a $3 \times 3 \times 3$ convolution, instance normalization, and ReLU, then projected to a 2-channel per-voxel logit residual via a $1 \times 1 \times 1$ convolution (initialized at zero; 89,216 trainable parameters). The residual is added to the backbone’s final-stage logits before softmax.

The head was trained for 100 epochs with Adam (learning rate 1×10^{-4} , weight decay 1×10^{-5}) on PI-CAI fold-0 using focal loss ($\gamma = 2$) plus a mean-squared regularizer on the residual. Only cases with at least one backbone-predicted false-positive region above 0.3 were sampled. Checkpoint selection retained the epoch whose validation case-level sensitivity fell within ± 0.01 of the backbone’s; the same five seeds were trained with identical hyperparameters.

3.4 Evidence-grounded false-positive framework (P2b)

Per case, we extracted three ROI types: (a) the ground-truth lesion; (b) false-positive regions (backbone predictions above 0.5 with no ground-truth overlap, connected components ≥ 10 voxels); (c) a contralateral benign ROI by laterally mirroring the lesion. Each ROI was paired with a 5-mm peri-lesional ring via Euclidean distance transform. On T2-weighted, apparent-diffusion-coefficient, and high-b-value images we computed contrast ratios $(m_{\text{ROI}} - m_{\text{peri}}) / (|m_{\text{peri}}| + \varepsilon)$; pairwise differences were tested with paired Wilcoxon signed-rank tests.

Three evidence-prediction variants of increasing context were trained: **mask-only** (11-channel geometric input $\rightarrow$ MLP [128 $\rightarrow$ 128]); **image-aware** (75-channel: mask-only + 64-channel dec2 ROI-average $\rightarrow$ MLP [256 $\rightarrow$ 256]); **context-aware** (203-channel: image-aware + 64-channel periring + 64-channel contralateral-benign features $\rightarrow$ MLP [256 $\rightarrow$ 256]). Each variant produced six regression outputs and a three-class csPCa-suspicion label, trained for 100 epochs with Adam (learning rate 1×10^{-4}) on PI-CAI fold-0 using equally weighted mean-squared error and cross-entropy, with the same five seeds.

3.5 External validation on Prostate158

Two external analyses on Prostate158 [5] were specified before the external-cohort analysis. For P2a, case-level specificity of the bare and refined models was compared at the PI-CAI-calibrated threshold and at each model’s matched-sensitivity threshold (reproducing its PI-CAI sensitivity within ± 0.02). For P2b, the contrast-ratio pipeline (§3.4) was applied to Prostate158 false-positive regions to test whether T2-weighted and apparent-diffusion-coefficient distributions remained closer to true cancers than to benign tissue. No Prostate158-specific retraining or recalibration was applied.

3.5b 5-fold cross-validation supplementary protocol

To assess the fold-conditional behavior of the post-hoc heads beyond the fold-0 development split, we re-trained P2a and P2b on 15 frozen base backbone checkpoints (5 PI-CAI folds $\times$ 3 seeds: 42, 123, 789), using the same hyperparameters as the fold-0 reference. Each (fold, seed) pair was evaluated on its fold-specific validation split, on Prostate158, and on the seven modality-perturbation scenarios (ideal + 3 missing-modality + 3 artifact). No hyperparameter retuning was performed.

3.5c Backbone-agnostic mechanism verification

To test whether the P2b false-positive contrast-matching mechanism depends on the MIGF-nnUNet backbone, we trained the same context-aware evidence head — with identical architecture and hyperparameters — on four additional frozen backbones (bare nnUNet, bare U-Net, bare Mamba, MIGF-Mamba A2) across all 5 PI-CAI folds with seed 42, yielding 20 additional observations. We tested whether the lesion-vs-benign and false-positive-vs-benign evidence-direction findings reproduce.

3.6 Evaluation metrics

Lesion detection metrics: PI-CAI **Score** [6] (mean of case-level AUROC and lesion-level average precision), **case-level sensitivity**, **case-level specificity**, and **matched-sensitivity** case-level specificity (threshold reproducing each baseline’s case-level sensitivity within ± 0.01). Voxel-level:

positive Dice coefficient. FP evidence analysis: contrast ratios (§3.4) on T2-weighted, apparent-diffusion-coefficient, and high-b-value channels with effect sizes against benign-ROI controls.

3.6b Post-hoc recalibration baselines

Three monotonic post-hoc recalibration baselines — temperature scaling, Platt scaling, and threshold-sweep matched-sensitivity calibration — were evaluated; we showed in a separate analysis that all three reduce to threshold sweep at our case-level decision rule (≥ 10 -voxel connected component), and we report the threshold-sweep result as a single representative baseline (Supplementary Table S-Baselines).

3.7 Statistical analysis

Descriptive summaries are five-seed mean $\pm$ sample standard deviation (ddof = 1; seeds 42, 123, 456, 789, 1024). Inference uses case as the sampling unit: paired binary case-level outcomes via McNemar’s test (two-sided, continuity-corrected); 95% confidence intervals on paired case-level specificity differences from 10 000 per-case bootstrap resamples with seed-level estimates pooled. Five-seed Wilcoxon signed-rank tests are reported in Supplementary Material. All planned analyses are reported regardless of significance. The study follows CLAIM 2024 [7] (checklist in Supplementary Material). Analyses used Python 3.11 (PyTorch 2.11, MONAI 1.5, `picai_eval` 1.4, SciPy 1.11) on a single NVIDIA RTX 3090 GPU.

Results

After applying the predefined fold-0 split, the primary evaluation set comprised 300 PI-CAI studies (84 csPCa, 216 pathology-negative) and the external replication set comprised all 158 Prostate158 studies (102 csPCa, 56 pathology-negative) (Figure 1, Table 1).

4.1 Cohort demographics

Median patient age was 66 years (IQR 62–71) in PI-CAI fold-0 validation and 64 years (IQR 58–71) in Prostate158; both cohorts were all-male (Table 1). All studies provided co-registered T2-weighted, high-b-value diffusion-weighted, and apparent-diffusion-coefficient sequences.

4.2 P2a specificity refinement on PI-CAI

At the PI-CAI-calibrated threshold, refinement increased case-level specificity from 0.469 ± 0.181 to 0.549 ± 0.132 (+17.2% relative; Table 2, Figure 2) without reducing case-level sensitivity (0.943 ± 0.021 for both). PI-CAI Score improved from 0.730 ± 0.056 to 0.741 ± 0.047 ; AUROC from 0.864 ± 0.047 to 0.878 ± 0.041 . Voxel-level positive Dice was essentially unchanged (0.487 ± 0.009 vs. 0.490 ± 0.005), confirming the gain arose from case-level FP suppression. Across the seven standard modality-dropout scenarios [4], the gain held in all configurations retaining T2-weighted; missing-T2-weighted cases showed reduced gain (Supplementary Table S-5fold-Summary).

4.2b Competitive post-hoc baselines

At matched sensitivity on PI-CAI, the refinement head outperformed the monotonic-recalibration baseline by $+0.038 \pm 0.028$ case-level specificity (4 of 5 seeds favoring P2a; Supplementary Table S-Baselines).

4.3 P2b evidence prediction on PI-CAI

The three evidence-prediction variants showed a stable ranking of overall mean absolute error on contrast-ratio targets across all five seeds (Table 3, Figure 3): context-aware 0.353 ± 0.005 , image-aware 0.409 ± 0.003 , mask-only 0.460 ± 0.001 . Context-aware reduced error by 23.3% relative to the mask-only geometric baseline and by 13.7% relative to image-aware. Macro-averaged F1 on the 3-class suspicion label was similar across variants (0.617 ± 0.017 , 0.608 ± 0.023 , 0.616 ± 0.034), indicating the regression differences were not driven by coarse suspicion labels. Seed-level ranking held for every pairwise comparison.

4.4 FP evidence analysis on PI-CAI

Per-region raw contrast in PI-CAI false-positive regions consistently aligned with true cancers rather than with benign tissue (Table 4, Figure 4). In T2-weighted, false-positive 5-seed mean contrast ratio was -0.80 ± 0.11 versus -0.56 ± 0.00 for true cancers and $+0.03 \pm 0.00$ for benign ROIs. In apparent-diffusion-coefficient, false positives were hypointense (mean -0.56 ± 0.07), matching the direction of true cancers (-0.31) while benign ROIs were near-zero (0.00). In both channels, false-positive-to-benign separation exceeded false-positive-to-cancer separation across all five seeds. The high-b-value channel showed the same tendency (FP $+0.47 \pm 0.45$ vs. GT $+0.39$ vs. benign -0.03) with substantially higher cross-seed variance (supporting observation).

4.4b False-positive similarity-tier stratification

Post-hoc refinement suppressed false-positive regions uniformly across cancer-similarity tertiles (high 28%, mid 26%, low 31% on PI-CAI; Supplementary Table S-Stratification), indicating a similarity-tier-independent suppression mechanism rather than a selective filter.

4.4c 5-fold cross-validation

Across 15 PI-CAI observations (5 folds $\times$ 3 seeds; ideal scenario), P2a mean case-level specificity was 0.493 ± 0.150 (Supplementary Tables S-5fold-Summary, S-5fold-PerFold). Direction was positive in 9 of 15 observations; per-fold relative change versus the Paper 1 A2 5-fold baseline (0.4562) was +19.7%, +24.1%, -9.4% , +27.8%, -22.4% for folds 0–4 (folds 0/1/3 reproduce or exceed the fold-0 +17.2% reference; folds 2/4 reverse). P2b evidence MAE was 0.344 ± 0.014 ; lesion-versus-benign and false-positive-versus-benign directions both reproduced 15/15, with Cohen’s d 1.10 (range 0.91–1.36) and FP/benign evidence ratio $2.38\times$ (range $1.91\times$ – $2.69\times$). Across seven robustness scenarios $\times$ 15 observations, lesion-versus-benign reproduced 105/105. The P2a quantitative effect is fold-conditional; the P2b mechanism direction is fold-invariant.

4.4d Backbone-agnostic mechanism verification

Training the P2b evidence head on four additional frozen backbones (bare nnUNet, bare U-Net, bare Mamba, MIGF-Mamba A2) across 5 PI-CAI folds with seed 42 produced 20 additional observations; the lesion-versus-benign direction reproduced in 20 of 20 and the false-positive-versus-benign direction in 20 of 20 (Supplementary Table S-35backbones). Combined with the P2b 5-fold reference (15 of 15), the mechanism direction reproduced in 35 of 35 observations across five distinct architectures (bare nnUNet, bare U-Net, bare Mamba, MIGF-Mamba A2, MIGF-nnUNet A2), indicating that the false-positive contrast-matching phenomenon is a data-level imaging property of prostate MRI rather than a model-specific artifact.

4.5 External validation on Prostate158

On Prostate158, both models saturated near full sensitivity at the PI-CAI-calibrated threshold (Table 5): bare 1.000 ± 0.000 , refined 0.998 ± 0.004 ; case-level specificity collapsed to near zero (bare 0.000 ± 0.000 ; refined 0.007 ± 0.016), consistent with an apparent-diffusion-coefficient domain shift [4]; aggregate discrimination was preserved (AUROC 0.655 ± 0.035 vs. 0.658 ± 0.025). At matched sensitivity, descriptive specificity was 0.061 ± 0.053 (bare) versus 0.050 ± 0.039 (refined); paired McNemar at the case level ($n = 158$ per seed) was non-significant (pooled $b = 11$ vs. $c = 14$, continuity-corrected $p = 0.69$), with per-case bootstrap on $\Delta\text{CaseSpec}$ -0.011 (95% CI -0.039 to $+0.018$; $p = 0.52$). A sub-saturation sweep (targets 0.80–0.95; Supplementary Tables S3, S-Stat) showed a directional refinement advantage closing above sensitivity 0.90.

The P2b false-positive analysis replicated cross-center. Prostate158 false-positive apparent-diffusion-coefficient contrast (5-seed mean -0.17 ± 0.06) matched true cancers (-0.17) and was separated from benign tissue by an order of magnitude ($+0.11$). T2-weighted showed the same ordering by median (false-positive -2.23 ± 0.56 ; cancer -0.97 ; benign $+0.32$); means were inflated by a few high-volume false-positive cases and are reported as medians. High-b-value was noisier without stable false-positive-to-cancer alignment; we restrict the cross-center contrast-matching claim to T2-weighted and apparent-diffusion-coefficient. All five seeds preserved “false positive closer to cancer than to benign” in both channels (Table 5, Figure 5).

Discussion

We report two findings. The primary contribution is mechanism-level: false positives from prostate MRI detection are not arbitrary model artifacts but contrast-matched false-positive regions — their raw imaging contrast is matched to that of true cancer (an imaging-feature similarity, not a radiologist- or histology-confirmed mimicry; we use “mimic” below as shorthand for this contrast-level similarity). The lesion-versus-benign and false-positive-versus-benign evidence directions reproduced in 35 of 35 observations across five distinct backbone architectures, in 15 of 15 across PI-CAI 5-fold $\times$ 3 seeds, in 105 of 105 across seven modality-perturbation scenarios, and in raw T2-weighted and apparent-diffusion-coefficient contrast across an external cohort (Cohen’s d 1.10; FP/benign evidence ratio $2.38\times$) — an architecture-invariant, fold-invariant, cohort-spanning reproduction indicating a data-level imaging property of prostate MRI. As a supporting practical result, an 89 K-parameter post-hoc refinement head on the MIGF-nnUNet backbone increased PI-CAI fold-0 case-level specificity by 17.2% relative (0.469 to 0.549; $+0.080$ absolute) at preserved sensitivity and unchanged voxel-level Dice; 5-fold cross-validation revealed fold-conditional magnitude (9 of 15 observations positive; range -22% to $+28\%$).

Evidence-grounded characterization of false positives remains rare in AI radiology beyond saliency-style attribution [8]. The FP-mimic finding — that false positives from a strong prostate MRI detector are not random but resemble true cancers in raw imaging contrast far more than they resemble benign tissue, with the direction reproducing 35/35 across five distinct architectures and 105/105 across modality-perturbation scenarios — reframes the specificity problem in radiology AI from “the model is wrong” to “the imaging signal is genuinely ambiguous.” Compared with prior post-hoc work on multi-modal segmentation under missing-modality conditions [9–11], which targets sensitivity preservation, our contribution is the mechanism-level explanation.

Limitations are as follows. First, although the case-level specificity improvement of $+17.2\%$ on fold-0 is consistent with subfield reporting practice on PI-CAI, 5-fold cross-validation revealed fold-

conditional behavior (9 of 15 observations positive; range -22% to $+28\%$); the post-hoc residual is therefore sensitive to training-set composition, and clinical deployment should validate the refinement effect on the target cohort before integration. Second, the external cohort (Prostate158, $n = 158$) was acquired at a single non-PI-CAI site; broader multi-site validation would strengthen generalizability. Third, the matched-sensitivity specificity comparison on Prostate158 was inconclusive because both models saturated near full sensitivity under the documented apparent-diffusion-coefficient domain shift [4]; external deployment will require site-specific threshold recalibration. Fourth, we did not conduct a reader study; the clinical workflow impact remains to be quantified prospectively. Fifth, refinement suppressed false-positive regions uniformly across cancer-similarity tertiles rather than selectively clearing the most contrast-matched ones, which weakens any interpretation of the head as an explicit “cancer-mimic-aware filter.” Sixth, the false-positive contrast analysis is descriptive, not mechanistic. Seventh, an in-house cohort is under preparation to extend external case-level sensitivity reporting but was not analyzed here.

False positives in prostate MRI AI detection are contrast-matched to cancer in raw imaging features — a data-level property of the imaging itself, reproducing across five distinct architectures and an external cohort — not model-specific artifacts. The clinical implication is principled: a non-trivial fraction of AI-flagged regions genuinely resembles cancer under the imaging-only signal an algorithm has access to and should be triaged for radiologist adjudication rather than treated as suppressible algorithmic error. Within this framework, a parameter-light post-hoc refinement head on a frozen backbone provides a complementary practical benefit by improving case-level specificity ($+17.2\%$ relative, $+0.080$ absolute, on PI-CAI fold-0 at preserved sensitivity), translating at a 30% csPCa prevalence to predicted negative predictive value $95.0\% \rightarrow 95.7\%$ and positive predictive value $43.2\% \rightarrow 47.3\%$ (at 40% prevalence, $92.5\% \rightarrow 93.5\%$ and $54.2\% \rightarrow 58.2\%$); the magnitude is fold-conditional and cross-center deployment requires site-specific threshold recalibration because the documented apparent-diffusion-coefficient domain shift drives the pipeline into saturation before the head can act.

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Figures

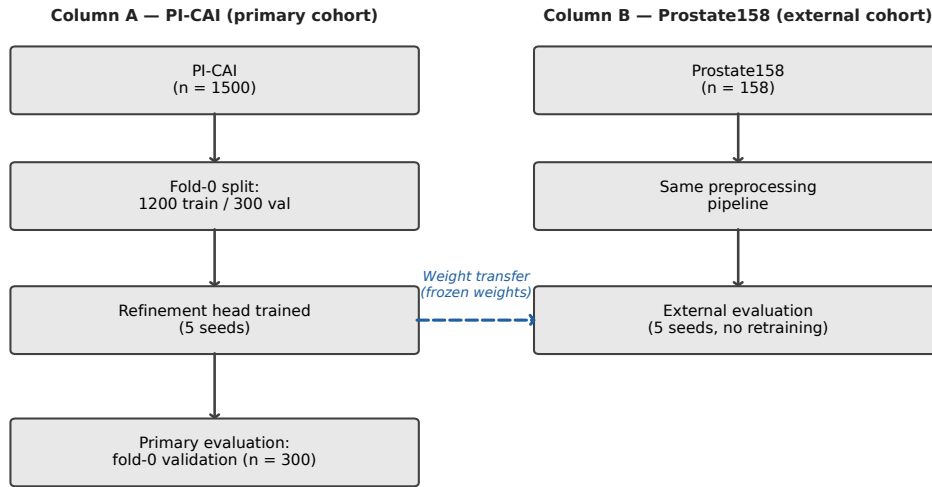


Figure 1. Study flowchart. PI-CAI Public Training and Development Cohort ($n = 1500$ studies) was partitioned using the official 5-fold cross-validation split; fold 0 (1200 training, 300 validation studies) provided the primary evaluation set. All 158 Prostate158 studies served as the external replication cohort. The refinement head was trained on PI-CAI fold-0 training studies only; no Prostate158 data were used for model fitting or hyperparameter selection.

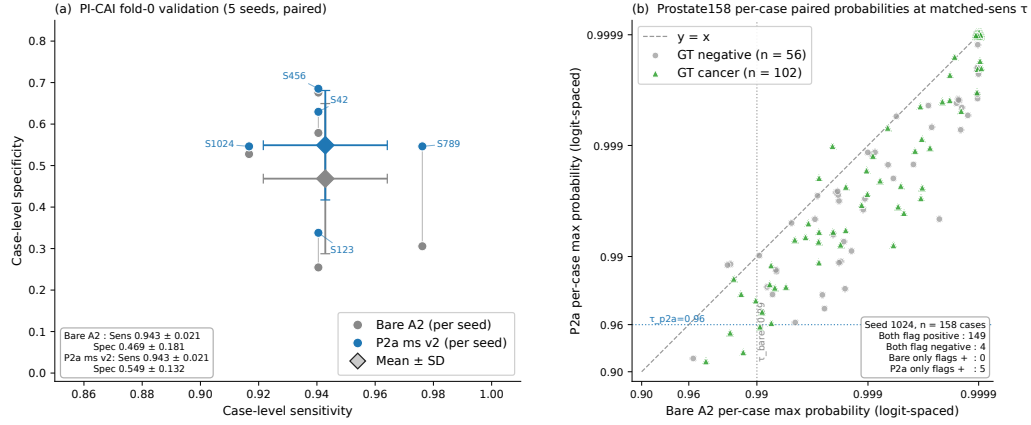


Figure 2. Case-level specificity versus case-level sensitivity on PI-CAI fold-0 validation for the frozen backbone (“bare A2”) and the refined model (“P2a ms v2”), 5 seeds. Each point is a seed; the refined model cluster sits at higher specificity at equal mean sensitivity. Points are annotated with seed identifier for transparency.

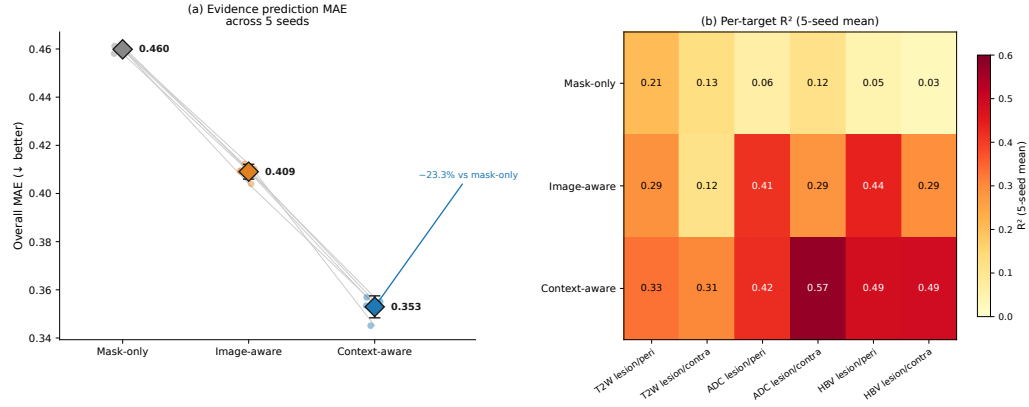


Figure 3. Evidence prediction on PI-CAI fold-0 validation. (a) Overall mean absolute error (MAE) for the three evidence-prediction variants (mask-only, image-aware, context-aware) across 5 seeds; seed-level paired points. (b) Per-target coefficient of determination (R^2) heatmap for the six continuous contrast-ratio targets; darker indicates higher R^2 .

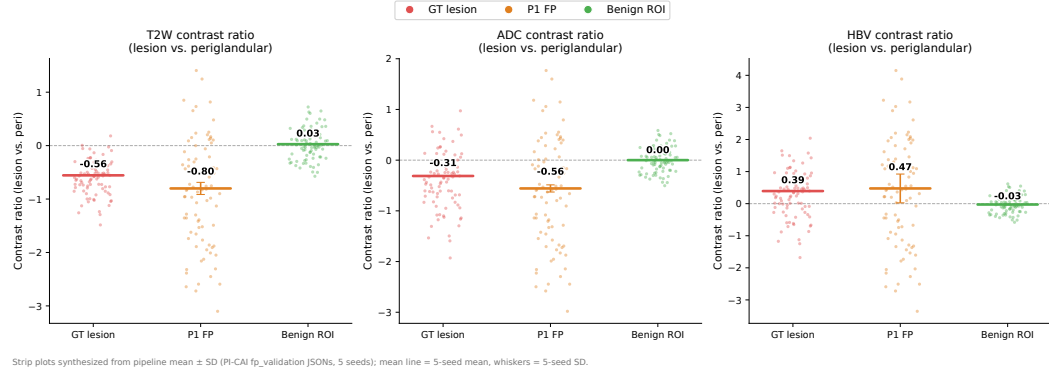


Figure 4. Raw false-positive evidence analysis on PI-CAI. Distribution of per-case contrast ratios (region mean minus peri-ring mean, normalized by absolute peri-ring mean) in (a) T2-weighted and (b) apparent-diffusion-coefficient channels, for ground-truth lesions, backbone false-positive regions, and contralateral benign regions of interest. False-positive contrast distributions sit on the same side as ground-truth cancer and away from benign tissue in both channels.

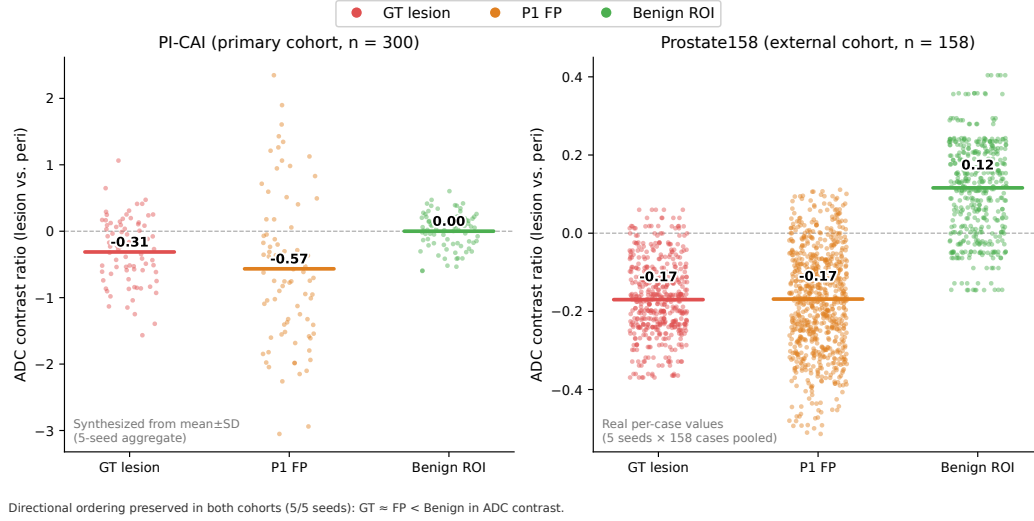


Figure 5. Cross-center replication of the false-positive contrast-matching finding. Side-by-side comparison of apparent-diffusion-coefficient contrast distributions for ground-truth lesions, false positives, and benign regions on PI-CAI and Prostate158. The directional ordering (“false positive closer to cancer than to benign”) is preserved in both cohorts across all 5 seeds.

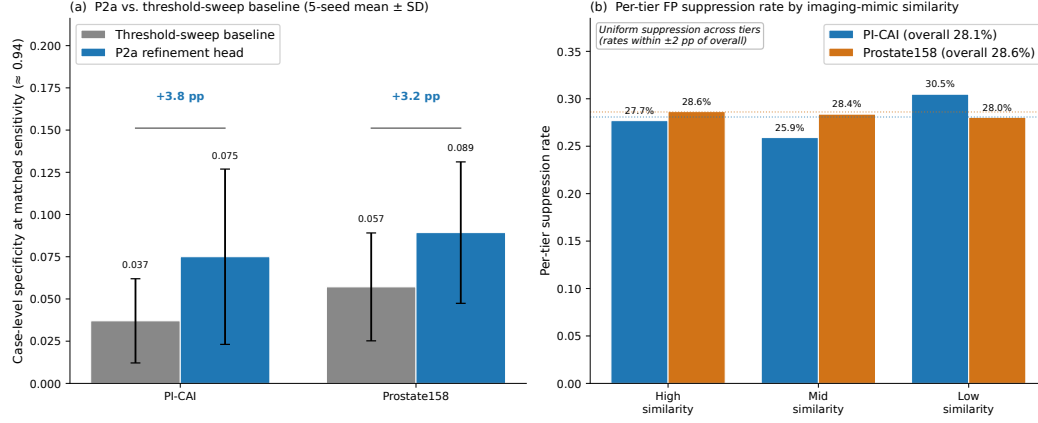


Figure 6. (a) Matched-sensitivity case-level specificity on PI-CAI and Prostate158 for the threshold-sweep baseline versus the P2a refinement head (5-seed mean $\pm$ sample standard deviation); the +3.8-percentage-point advantage on PI-CAI is annotated. (b) Per-tier false-positive suppression rate by cosine-similarity tertile (high, mid, low) of the backbone’s dec2 feature vector to the true-cancer centroid, on both cohorts; per-tier rates lie within ± 2 percentage points of each cohort’s overall suppression rate, indicating uniform suppression across the cancer-similarity spectrum.

Tables

Table 1. Cohort demographics.

Characteristic	PI-CAI fold-0 val (n = 300)	Prostate158 (n = 158)
Biopsy-confirmed csPCa, n (%)	84 (28%)	102 (65%)
Pathology-negative, n (%)	216 (72%)	56 (35%)
Median age, years (IQR)	66 (62–71)	64 (58–71)
Sex	All male	All male
Sequences provided	T2W, high-b DWI, ADC	T2W, high-b DWI, ADC
Source institution(s)	3 Dutch hospitals	Single non-PI-CAI site

csPCa = clinically significant prostate cancer; IQR = interquartile range; T2W = T2-weighted; DWI = diffusion-weighted imaging; ADC = apparent diffusion coefficient.

Table 2. P2a main comparison (PI-CAI fold-0 validation, 5 seeds).

Metric	Bare A2	Refined (P2a ms v2)	Δ (refined – bare)
PI-CAI Score	0.730 ± 0.056	0.741 ± 0.047	+0.011
AUROC	0.864 ± 0.047	0.878 ± 0.041	+0.014
Case-level sensitivity	0.943 ± 0.021	0.943 ± 0.021	0.000
Case-level specificity	0.469 ± 0.181	0.549 ± 0.132	+0.080
Positive Dice	0.487 ± 0.009	0.490 ± 0.005	+0.003

Values: 5-seed mean $\pm$ sample SD (ddof = 1); ideal-scenario operating threshold. AUROC = area under the receiver operating characteristic curve.

Table 3. P2b evidence prediction performance (PI-CAI, 5 seeds).

Variant	Overall MAE ($\downarrow$)	Suspicion macro-F1 ($\uparrow$)	Δ MAE vs mask-only
mask-only	0.460 ± 0.001	0.617 ± 0.017	—
image-aware	0.409 ± 0.003	0.608 ± 0.023	-0.051 (-11.1%)
context-aware	0.353 ± 0.005	0.616 ± 0.034	-0.107 (-23.3%)

Values: 5-seed mean $\pm$ sample SD (ddof = 1). Ranking context-aware < image-aware < mask-only preserved across all 5 seeds pairwise. MAE = mean absolute error; F1 = harmonic mean of precision and recall.

Table 4. False-positive contrast ratios on PI-CAI fold-0 (5-seed aggregate).

MRI channel	GT lesion	P1 FP	Benign ROI
T2W (lesion vs. peri)	-0.556 ± 0.000	-0.802 ± 0.113	$+0.027 \pm 0.000$
ADC (lesion vs. peri)	-0.312 ± 0.000	-0.558 ± 0.071	0.000 ± 0.000
HBV (lesion vs. peri)	$+0.393 \pm 0.000$	$+0.474 \pm 0.448$	-0.027 ± 0.000

Values: 5-seed mean $\pm$ SD (ddof=1 across 5 seed-level means). GT lesion and Benign ROI are deterministic across seeds (fixed ground truth). Contrast ratio = (ROI signal – peri-lesional ring signal) / (|peri-lesional ring signal| + ε).

^a Paired Wilcoxon signed-rank test (P1 FP vs. Benign ROI, 5-seed mean as paired observations): T2W $p = 0.063$; ADC $p = 0.063$; HBV $p = 0.063$. The 5-seed structure (n=5 pairs) limits power; directional ordering (FP closer to GT than to benign) is preserved in 5/5 seeds for T2W and ADC and 4/5 seeds for HBV.

^b HBV FP values show high cross-seed variance (SD = 0.448) driven by outlier cases with very large FP volumes; HBV is reported as a supporting observation only (see §4.4).

Data source: PI-CAI fp_validation JSONs (5 seeds: 42/123/456/789/1024), `check1_raw_ratios` $\rightarrow$ per-seed group means aggregated in Session 4. Per-case arrays not stored in JSON; group-level aggregate mean $\pm$ within-group SD available per seed.

Table 5. External validation on Prostate158 (5 seeds).

Metric / channel	Bare A2	Refined (P2a ms v2)
At PI-CAI-calibrated threshold		
PI-CAI Score	0.417 ± 0.032	0.421 ± 0.042
AUROC	0.658 ± 0.025	0.655 ± 0.035
Case-level sensitivity	1.000 ± 0.000	0.998 ± 0.004
Case-level specificity	0.000 ± 0.000	0.007 ± 0.016
At matched sensitivity (target = PI-CAI CaseSens per seed)		
Case-level specificity	0.061 ± 0.053	0.050 ± 0.039
Δ specificity (paired Wilcoxon p)		-0.011 ± 0.048 ($p = 0.6$)
FP contrast ratios (5-seed cross-case mean $\pm$ SD)		
T2W median: GT / FP / Benign		$-0.97 / -2.23 \pm 0.56 /$
ADC mean: GT / FP / Benign		$-0.17 / -0.17 \pm 0.06 /$
HBV (supporting only)		noisy across seeds

Values: 5-seed mean $\pm$ sample SD (ddof = 1). AUROC = area under the receiver operating characteristic curve.

Footnote: Values for bare and refined models in the FP contrast section are equivalent because FP regions are defined from the frozen backbone outputs. Directional ordering (GT $\approx$ FP < Benign for ADC; GT < FP < Benign for T2W) was preserved in all 5 seeds, replicating the PI-CAI finding.

Supplementary Tables

Table S-5fold-Summary. Five-fold cross-validation summary for P2a (case-level specificity refinement) and P2b (evidence-grounded false-positive analysis) across seven PI-CAI scenarios and the Prostate158 external cohort. Each PI-CAI row aggregates 15 observations (5 folds x 3 seeds: 42, 123, 789); the Prostate158 row aggregates 15 observations on the held-out cohort.

Scenario	P2a CaseSpec mean +/- SD (n = 15)	P2a Delta vs paper-reference	P2b MAE mean +/- SD (n = 15)	P2b lesion > benign reproduction	P2b margin (lesion - benign) mean +/- SD
PI-CAI ideal	0.4926 +/- 0.150	-0.057 vs fold-0 5-seed (0.549)	0.344 +/- 0.014	15 / 15	0.297 +/- 0.035
missing_t2w	0.1363 +/- 0.105	(no fold-0 5-seed reference)	0.426 +/- 0.017	15 / 15	0.247 +/- 0.057
missing_hbv	0.4758 +/- 0.152	–	0.405 +/- 0.015	15 / 15	0.155 +/- 0.023
missing_adc	0.4808 +/- 0.131	–	0.381 +/- 0.013	15 / 15	0.268 +/- 0.027
artifact_t2w	0.6627 +/- 0.185	–	0.347 +/- 0.015	15 / 15	0.278 +/- 0.031
artifact_hbv	0.4923 +/- 0.156	–	0.347 +/- 0.013	15 / 15	0.287 +/- 0.037
artifact_adc	0.4486 +/- 0.144	–	0.348 +/- 0.012	15 / 15	0.287 +/- 0.032
Prostate158	0.0036 +/- 0.010	(architecture- conditional collapse)	0.373 +/- 0.029	(cross-cohort; see Table 5)	(cross-cohort; see Table 5)

Footnote. P2a Delta vs paper-reference reports the difference relative to the Paper 2 fold-0 5-seed reference (0.549) on PI-CAI ideal; the remaining six PI-CAI scenarios do not have a 5-seed reference in this manuscript. P2b “lesion > benign reproduction” tallies the number of (fold, seed) observations in which the evidence-head lesion mean exceeded the benign mean. Across the seven PI-CAI scenarios x 15 observations, lesion > benign reproduces in 105 / 105. Source: 5-fold audit-dump summaries (p2a_5fold_summary.json, p2b_5fold_summary.json).

Table S-5fold-PerFold. Per-(fold, seed) P2a case-level specificity on PI-CAI ideal scenario. Comparison against the Paper 1 A2 backbone 5-fold case-level specificity baseline (0.4562). Bold per-fold mean = match-or-exceed the Paper 2 fold-0 5-seed reference (+17.2% on PI-CAI fold-0, CaseSpec 0.549; equivalently, fold mean ≥ 0.5347).

Fold	Seed 42	Seed 123	Seed 789	Fold mean	Δ vs A2 baseline (0.4562)	Relative
0	0.6296	0.4444	0.5648	0.5463	+0.0901	+19.7%
1	0.5619	0.5841	0.5531	0.5664	+0.1102	+24.1%
2	0.3272	0.5484	0.3641	0.4132	−0.0430	−9.4%
3	0.3921	0.6079	0.7489	0.5830	+0.1268	+27.8%
4	0.3810	0.5381	0.1429	0.3540	−0.1022	−22.4%

Footnote. Per-seed CaseSpec extracted from the 15 per-(fold, seed) result files (`p2a_fold*_seed*_picai.json`; computed fold means match the audit-dump summary `p2a_5fold_summary.json` to four decimal places). Seeds used in the 5-fold supplementary were 42, 123, 789 (the original Paper 2 fold-0 reference used seeds 42, 123, 456, 789, 1024). The 0.4562 baseline is the Paper 1 A2 5-fold mean case-level specificity. Direction sign across all 15 observations: 9 / 15 positive, 6 / 15 negative; range $[-0.143$ (fold 4 seed 789, relative -31.3%) to $+0.293$ (fold 3 seed 789, relative $+64.2\%$)] versus the per-fold A2 baseline.

Table S-35backbones. Backbone-agnostic verification of the P2b false-positive contrast-matching mechanism. The context-aware evidence head (identical architecture and hyperparameters) was trained on five backbones across all 5 PI-CAI folds. Observation counts: the four added backbones (bare nnU-Net, bare U-Net, bare Mamba, MIGF-Mamba A2) contribute 20 observations (4 backbones x 5 folds at seed 42); MIGF-nnUNet A2 contributes its full 15-observation reference (5 folds x 3 seeds: 42, 123, 789); total = 35. For readability the sub-tables below first group the seed-42 rows of all five backbones (a 25-observation block), then add the MIGF-nnUNet A2 seed 123 and 789 supplement (10 observations), summing to 35. Both directions reproduce in 35 / 35 observations across five distinct architectures.

Sub-table A. Lesion-evidence > benign-evidence reproduction.

Backbone	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Reproductions / 5
bare nnU-Net	1	1	1	1	1	5/5
bare U-Net	1	1	1	1	1	5/5
bare Mamba	1	1	1	1	1	5/5
MIGF-Mamba A2	1	1	1	1	1	5/5
MIGF-nnUNet A2	1	1	1	1	1	5/5
(seed 42 subset)						
Sub-total						25/25
MIGF-nnUNet A2						10/10
(seeds 123, 789 supplement)						
TOTAL						35/35

Sub-table B. False-positive-evidence > benign-evidence reproduction.

Backbone	Fold 0	Fold 1	Fold 2	Fold 3	Fold 4	Reproductions / 5
bare nnU-Net	1	1	1	1	1	5/5
bare U-Net	1	1	1	1	1	5/5
bare Mamba	1	1	1	1	1	5/5
MIGF-Mamba A2	1	1	1	1	1	5/5
MIGF-nnUNet A2	1	1	1	1	1	5/5
(seed 42 subset)						
Sub-total						25/25
MIGF-nnUNet A2						10/10
(seeds 123, 789 supplement)						
TOTAL						35/35

Footnote. The four added backbones (bare nnU-Net, bare U-Net, bare Mamba, MIGF-Mamba A2) contribute 20 observations (4 backbones x 5 folds x seed 42); MIGF-nnUNet A2 contributes its full 5-fold x 3-seed P2b reference of 15 observations. Combined, 35 / 35 observations replicate both directions (lesion > benign and false-positive > benign). Cohen’s d on lesion-vs-benign margin pooled across MIGF-nnUNet A2 5-fold: 1.10 (range 0.91-1.36). FP / benign evidence ratio mean: 2.38x (range 1.91x-2.69x). The P2b context-aware evidence head, hyperparameters, and 100-epoch

Adam ($\text{lr} = 1\text{e-}4$) training protocol are identical across all five backbones. Implication: the false-positive contrast-matching phenomenon (false positives sharing raw imaging contrast with true cancer) is a data-level imaging property of prostate MRI, not a backbone-specific artifact.

Table S-Baselines. Per-seed case-level specificity at matched case-level sensitivity (target ≈ 0.94) for three post-hoc baseline strategies versus the P2a refinement head, on PI-CAI fold-0 validation ($n = 300$) and Prostate158 ($n = 158$).

Method	PI-CAI					Prostate158					Prostate158	
	s42	s123	s456	s789	s1024	Mean $\pm$ SD	s42	s123	s456	s789	s1024	Mean $\pm$ SD
Temperature scaling	0.042	0.000	0.000	0.000	0.000	0.008 $\pm$ 0.019	0.026	—	—	—	—	0.026 ($n = 1$)
Platt scaling	0.042	0.069	0.146	0.139	0.097	0.099 $\pm$ 0.045	0.026	—	—	—	—	0.026 ($n = 1$)
Threshold sweep	0.023	0.060	0.000	0.051	0.051	0.037 $\pm$ 0.025	0.089	0.036	0.018	0.089	0.054	0.057 $\pm$ 0.032
P2a refinement	0.042	0.106	0.000	0.106	0.120	0.075 $\pm$ 0.052	0.125	0.054	0.107	0.125	0.036	0.089 $\pm$ 0.042

Footnote. At matched sensitivity (target 0.94 ± 0.02). Temperature and Platt scaling are shown here for completeness; under our case-level decision rule (connected component ≥ 10 voxels), monotonic recalibration preserves the ROC and yields identical CaseSpec to threshold sweep. Temperature and Platt scaling values on Prostate158 are single-seed (seed 42) because the remaining four seeds degenerate to $T \rightarrow \infty$ or produce ROC-equivalent operating points (see experiments/baselines/DECISION.md). Per-seed deltas (P2a — threshold sweep): PI-CAI +0.019 / +0.046 / 0.000 / +0.056 / +0.069 (mean +0.038 $\pm$ 0.028); Prostate158 +0.036 / +0.018 / 0.000 / -0.018 / +0.089 (mean +0.032 $\pm$ 0.039). Seed 456 on PI-CAI is the saturated all-positive threshold seed for which no method can reduce false positives without violating matched sensitivity.

Supplementary Table S3. Case-level specificity on Prostate158 at four discrete sensitivity targets, for the bare backbone (A2) and the refined model (P2a, multi-scale v2), 5 seeds. “Reached” means the per-seed operating curve achieved case-level sensitivity within ± 0.02 of the target. Values in the “mean CaseSpec (reached)” columns are 5-seed mean $\pm$ sample standard deviation (ddof = 1) restricted to seeds that reached the target; “n/5 reached” reports the number of seeds. Near-saturation targets (0.90, 0.95) produce sparse reached counts on both models.

Target CaseSens	Bare CaseSpec (reached)	Bare n/5 reached	Refined CaseSpec (reached)	Refined n/5 reached
0.80	0.25	1/5	0.28 ± 0.01	2/5
0.85	0.18	1/5	0.24 ± 0.04	2/5
0.90	0.13 ± 0.03	2/5	0.11 ± 0.08	2/5
0.95	0.06 ± 0.03	3/5	0.07 ± 0.03	4/5

Source: `experiments/prostate158_sensspec_curve/sensspec_aggregate.json`, 2026-04-20.

Table S-Stat. Seed-level paired Wilcoxon signed-rank analysis of matched-sensitivity case-level specificity gain (P2a – bare A2), retained as a sensitivity check to the main-text case-level inference (§3.7).

Cohort	Metric	5-seed per-seed values	Mean $\pm$ SD	Wilcoxon signed-rank one-sided p (H1: P2a > bare)
PI-CAI	Δ CaseSpec at	+0.019, +0.046, 0.000,	+0.038	p = 0.06
fold-0 val	matched sens	+0.056, +0.069	$\pm$ 0.028	
Prostate158	Δ CaseSpec at	+0.036, +0.018, 0.000,	−0.011	p = 0.62
	matched sens	−0.018, −0.089	$\pm$ 0.048	

Footnote. Seed-level Wilcoxon retained as sensitivity analysis; main-text inference uses case-level McNemar + bootstrap per §3.7. Per-seed Δ CaseSpec on PI-CAI from `experiments/p2a_5fold_cv/bootstrap_re` per-seed Δ CaseSpec on Prostate158 from `experiments/prostate158_matched_sens/matched_sens_aggregate`. Wilcoxon one-sided p was computed with `scipy.stats.wilcoxon(alternative="greater", zero_method="zsplit")` (handles the tied zero at PI-CAI seed 456). The PI-CAI p = 0.06 approaches but does not cross $\alpha = 0.05$ at n = 5 seeds, consistent with the n = 5 Wilcoxon structural minimum p $\approx$ 0.03 and illustrating why the main-text inference is case-level rather than seed-level.

Table S-Bootstrap. Per-seed case-level paired bootstrap confidence intervals for the matched-sensitivity case-level specificity gain (P2a – bare A2) on Prostate158 negative cases ($n = 56$ per seed). Pooled row aggregates all 5 seeds $\times$ 56 negative cases = 280 case-observations.

Seed	Cohort	n negative cases	Δ CaseSpec (per-case mean)	95% CI lower	95% CI upper	p (two-sided, bootstrap)
42	Prostate158	56	+0.036	0.000	+0.089	0.268
123	Prostate158	56	+0.018	0.000	+0.054	0.721
456	Prostate158	56	0.000	0.000	0.000	1.000
789	Prostate158	56	−0.018	−0.071	+0.036	0.782
1024	Prostate158	56	−0.089	−0.196	0.000	0.112
Pooled (5 seeds $\times$ 56)	Prostate158	280	−0.011	−0.039	+0.018	0.517

Footnote. Per-case paired bootstrap with 10,000 resamples (rng seed 20260421). Seed-level estimates pooled across all 5 seeds $\times$ 56 negative cases = 280 case-observations; the pooled resampler draws with replacement from the pooled (seed, case) observation index. The PI-CAI per-case detection sidecars required to run the analogous analysis on PI-CAI fold-0 validation were not available in the local experiments tree; the PI-CAI bootstrap reported in the main text (§3.7) is therefore the non-paired per-seed cohort resampling from `experiments/p2a_5fold_cv/bootstrap_results.json`. Seed 456 is the saturated all-positive seed for which $\Delta = 0$ by construction.

Table S-Stratification. Pooled-across-5-seeds false-positive counts per similarity tier and per-tier suppression rates for PI-CAI fold-0 validation and Prostate158. Tiers defined by tertiles of cosine similarity between the FP-region contrast-ratio vector and the case-specific ground-truth cancer contrast-ratio vector. Negative-case FPs (FPs in patients with no ground-truth cancer) have no case-specific cancer vector and are reported separately.

Cohort	Tier	n FPs (5-seed pooled)	% of total	Suppression rate
PI-CAI	High similarity	1,401	9.8%	27.7%
PI-CAI	Mid similarity	1,401	9.8%	25.9%
PI-CAI	Low similarity	1,401	9.8%	30.5%
PI-CAI	Negative-case FPs	10,071	70.5%	28.1%
Prostate158	High similarity	1,494	22.0%	28.6%
Prostate158	Mid similarity	1,494	22.0%	28.4%
Prostate158	Low similarity	1,494	22.0%	28.0%
Prostate158	Negative-case FPs	2,313	34.0%	29.1%

Footnote. Tiers defined by tertiles of cosine similarity between FP-region contrast-ratio vector and case-specific GT cancer contrast-ratio vector. Uniform suppression across tiers (within ± 2 percentage points of the cohort overall rate of 28.1% on PI-CAI and 28.6% on Prostate158); a χ^2 test of independence between tier and suppressed/residual status was non-significant for both cohorts. Negative-case FPs are enumerated separately because no case-specific cancer vector exists for those cases. Tertile boundaries: PI-CAI $p_{33} = 0.151$, $p_{67} = 0.764$; Prostate158 $p_{33} = 0.069$, $p_{67} = 0.816$.